

Floer Homology and Knot Concordance

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Transcribed from three-page handwritten notes. [Download the printable transcript \(PDF\)](#). The knot, crossing, and graph diagrams are described in words alongside the matrices and formulas.

Page 1: Ribbon disks and obstructions to sliceness

Consider the knot 6_1 , drawn in the notes as a two-loop knot diagram with a clasp and a twisted band. This bounds a disk, but not an embedded one.

A **ribbon disk** is an immersed disk in S^3 whose singularities are ribbon singularities. The local picture shows a band passing through the interior of another sheet of the disk. The double-point arc has two preimage arcs: one with endpoints on the disk boundary, and one in the interior.

Since S^3 is the boundary of D^4 , one can push parts of a ribbon disk from S^3 into $\text{int}(D^4)$ to obtain a smoothly embedded disk whose boundary is the original knot. A knot $K \subset S^3$ which bounds a smoothly embedded disk in D^4 is called a **slice knot**.

Classical obstructions to sliceness

If K is a slice knot:

- $\Delta_K(t) = f(t)f(t^{-1})$ for some $f(t) \in \mathbb{Z}[t]$.
- $\det(K) = |\Delta_K(-1)| = f(-1)^2$.
- $\sigma(K) = 0$.
- $\tau(K) = 0$ (knot Floer homology).
- $s(K) = 0$ (Khovanov homology).
- $\delta(K) = 0$ (Floer homology of $\Sigma(K)$).

Theorem (Casson–Gordon). If K is slice, then $\Sigma(K) = \partial W^4$, where W^4 is a rational homology ball.

Definition. $\Sigma(K)$ is the branched double cover of S^3 , with branch locus mapping to K .

Constructing the branched double cover

Start from a knot diagram and checkerboard-color its complementary regions. Choose a marked colored region. The notes draw a graph with one vertex for each colored region and an edge for each crossing.

In the illustrated example the graph has three vertices. The left and right vertices are connected by two positive edges; the left vertex is connected to the bottom marked vertex by a negative edge; and the right vertex is connected to the marked vertex by a positive edge.

Page 2: The graph matrix and Donaldson's theorem

The local sign convention is illustrated by two crossings with their left and right sectors colored. When the strand from southwest to northeast passes over the other strand, the crossing contributes a positive edge; switching the overpassing strand gives a negative edge.

Discard the marked vertex. Label each remaining vertex by the negative sum of its incident crossing signs in the unreduced graph. The illustrated **reduced graph** has two vertices labeled -1 and -3 , joined by two positive edges.

Fact. The reduced graph encodes K , but not quite its diagram.

To obtain $\Sigma(K)$, put an unknot at every vertex and a clasp for every edge. The diagram has two unknots, with framings -1 and -3 , joined by two clasps. Surgery on this framed link gives $\Sigma(K)$.

From the graph, get a matrix: diagonal entries are the labels at vertices; off-diagonal entries are the signed sums of the edges between vertices. In the example,

$$G_{\Gamma} = \begin{pmatrix} -1 & 2 \\ 2 & -3 \end{pmatrix}.$$

This is the linking matrix of the associated framed link and the intersection matrix of a 4-manifold X_{Γ} . Starting with a diagram for K , this constructs X_{Γ} with boundary $\Sigma(K)$.

If one starts with an alternating diagram for K , then G_{Γ} is negative definite, and so is X_{Γ} .

Theorem A (Donaldson). If X is a closed, orientable, negative-definite, smooth 4-manifold, its intersection form is diagonalizable.

If K is slice, then $X_{\Gamma} \cup W^4$ is a closed, negative-definite, oriented 4-manifold. Consequently G_{Γ} is represented by $-I_r$, where $r = \text{rank}(G_{\Gamma})$. Concretely, there exists a matrix $A \in M_r(\mathbb{Z})$ such that

$$G_{\Gamma} = -AA^T.$$

If no such factorization exists, K is not slice.

This shows that $K = P(-3, 5, 7)$ is not smoothly slice. On the other hand, $\Delta_K = 1$ implies that K is topologically slice (Freedman).

Page 3: Heegaard Floer homology and examples

More recently, this method was used by Lisca (2007) to classify slice 2-bridge knots.

Another test: Heegaard Floer homology

Ozsváth and Szabó defined a d -invariant, written in the notes as

$$d : H^2(Y^3) \rightarrow \mathbb{Q}.$$

If $Y^3 = \Sigma(K)$ and K is slice, then d vanishes on a special subgroup of H^2 .

Theorem. Let K be a slice knot and D a diagram which is alternating or whose reduced graph is a tree, with G_Γ negative definite. Then:

- $G_\Gamma = -AA^T$.
- Every class in $\text{coker}(A^T)$ has a representative all of whose entries are ± 1 .

The example 6_1

The notes checkerboard-color a diagram of 6_1 . Its reduced graph is a single positive edge between vertices of weights -2 and -5 . Thus

$$G_\Gamma = \begin{pmatrix} -2 & 1 \\ 1 & -5 \end{pmatrix} = - \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}.$$

The zero class has the representative

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \in \text{im}(A^T).$$

The two other displayed representatives are

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

The cokernel is $\mathbb{Z}/3$, and $|\det(A^T)| = |\text{coker}(A^T)| = 3$.

With more twists, say $n \geq 5$, the illustrated graph still has one positive edge, with vertex weights -2 and $-(n+1)$. Its matrix is

$$G_\Gamma = \begin{pmatrix} -2 & 1 \\ 1 & -(n+1) \end{pmatrix}.$$

The required integral factorization would imply

$$\det(A^T) = \sqrt{2n+1} \in \mathbb{Z}.$$

For $n \geq 5$, the notes point out that this integer is at least 4, the number of vectors with two entries each equal to ± 1 .

The final drawing is the three-stranded pretzel knot $P(-3, 5, 7)$: three successive twisted bands carry parameters -3 , 5 , and 7 , with their ends joined in the usual pretzel pattern. This is the knot mentioned on page 2 as topologically but not smoothly slice.